

Rocket-based combined cycle

The **RBCC**, or rocket-based combined cycle propulsion system, was one of the two types of propulsion systems that may have been tested in the Boeing X-43 experimental aircraft. The RBCC, or strutjet as it is sometimes called, is a combination propulsion system that consists of a ramjet, scramjet, and ducted rocket, where all three systems use a shared flow path.^[1]

A **TBCC**, or turbine-based combined cycle propulsion system, is a turbine engine combined with a ramjet and scramjet.^[2]

A **TRCC**, or turbo rocket combined cycle propulsion system, is another combination propulsion system that combines an afterburning turbine engine with a RBCC propulsion system.^[3]

See also

- SABRE (Synergistic Air Breathing Rocket Engine), a pre-cooled air-breathing rocket/RAM-jet engine based on General Dynamics' exploration of LACE concepts (Liquid Air Cycle Engine) by Reaction Engines, UK.^[4]

References

1. Sketch of a RBCC propulsion system (scroll down to 7th picture) (<https://web.archive.org/web/20070211045732/http://www.affordablespaceflight.com/nasa2.html>)
2. Sketch of a TBCC propulsion system (scroll down to 5th picture) (<https://web.archive.org/web/20070211045732/http://www.affordablespaceflight.com/nasa2.html>)
3. Sketch of a TRCC propulsion system (scroll down to 9th picture) (<https://web.archive.org/web/20070212020658/http://www.affordablespaceflight.com/nasa1.html>)
4. "First Sabre Development Milestone Reached" (https://web.archive.org/web/20150402150244/http://www.reactionengines.co.uk/press_release/3a_Milestone_Release_PUBLIC27Jan2015.pdf) (PDF). January 26, 2015. Archived from the original (http://www.reactionengines.co.uk/press_release/3a_Milestone_Release_PUBLIC27Jan2015.pdf) (PDF) on 2015-04-02. Retrieved 2015-02-27.

External links

- Performance Evaluation of the NASA GTX RBCC Flowpath (<https://web.archive.org/web/20110610051911/http://gltrs.grc.nasa.gov/reports/2001/TM-2001-210953.pdf>) - Glenn Research Center - NASA
- Parametric Study Conducted of Rocket-Based, Combined-Cycle Nozzles (<https://web.archive.org/web/20080321075958/http://www.grc.nasa.gov/WWW/RT/RT1997/5000/5880steffen.htm>) - Glenn Research Center - NASA
- Aerojet Successfully Tests RBCC Single Thruster, Demonstrating Tri-Fluid Rocket Injector Capabilities (<http://www.spaceref.com/news/viewpr.html?pid=11878>) - SpaceRef

- Hypersonic inlet studies for a small-scale rocket-based combined-cycle engine (<http://cat.inist.fr/?aModele=afficheN&cpsidt=19683011>), Journal of propulsion and power, 2007, vol. 23, no6, pp. 1160–1167, AIAA.
- Rocket-Based Combined-Cycle Engine (RBCC): Ramrocket (<https://web.archive.org/web/20110514154624/http://caius.utias.utoronto.ca/rbcc.html>), University of Toronto, High-Speed Vehicle Propulsion Systems Group.

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